



Olaparib combined with abiraterone in patients with metastatic castration-resistant prostate cancer: a randomised, double-blind, placebo-controlled, phase 2 trial

Noel Clarke, Pawel Wiechno, Boris Alekseev, Nuria Sala, Robert Jones, Ivo Kocak, Vincenzo Emanuele Chiuri, Jacek Jassem, Aude Fléchon, Charles Redfern, Carsten Goessl, Joseph Burgents, Robert Kozarski, Darren Hodgson, Maria Learoyd, Fred Saad

Summary

Background Patients with metastatic castration-resistant prostate cancer and homologous recombination repair (HRR) mutations have a better response to treatment with the poly(ADP-ribose) polymerase inhibitor olaparib than patients without HRR mutations. Preclinical data suggest synergy between olaparib and androgen pathway inhibitors. We aimed to assess the efficacy of olaparib plus the androgen pathway inhibitor abiraterone in patients with metastatic castration-resistant prostate cancer regardless of HRR mutation status.

Methods We carried out this double-blind, randomised, placebo-controlled phase 2 trial at 41 urological oncology sites in 11 countries across Europe and North America. Eligible male patients were aged 18 years or older with metastatic castration-resistant prostate cancer who had previously received docetaxel and were candidates for abiraterone treatment. Patients were excluded if they had received more than two previous lines of chemotherapy, or had previous exposure to second-generation antihormonal drugs. Patients were randomly assigned (1:1) using an interactive voice or web response system, without stratification, to receive oral olaparib 300 mg twice daily or placebo. All patients received oral abiraterone 1000 mg once daily and prednisone or prednisolone 5 mg twice daily. Patients and investigators were masked to treatment allocation. The primary endpoint was investigator-assessed radiographic progression-free survival (rPFS; based on Response Evaluation Criteria in Solid Tumors version 1.1 and Prostate Cancer Clinical Trials Working Group 2 criteria). Efficacy analyses were done in the intention-to-treat population, which included all randomly assigned patients, and safety analyses included all patients who received at least one dose of olaparib or placebo. This trial is registered with ClinicalTrials.gov, number NCT01972217, and is no longer recruiting patients.

Findings Between Nov 25, 2014, and July 14, 2015, 171 patients were assessed for eligibility. Of those, 142 patients were randomly assigned to receive olaparib and abiraterone (n=71) or placebo and abiraterone (n=71). The clinical cutoff date for the final analysis was Sept 22, 2017. Median rPFS was 13·8 months (95% CI 10·8–20·4) with olaparib and abiraterone and 8·2 months (5·5–9·7) with placebo and abiraterone (hazard ratio [HR] 0·65, 95% CI 0·44–0·97, p=0·034). The most common grade 1–2 adverse events were nausea (26 [37%] patients in the olaparib group vs 13 [18%] patients in the placebo group), constipation (18 [25%] vs eight [11%]), and back pain (17 [24%] vs 13 [18%]). 38 (54%) of 71 patients in the olaparib and abiraterone group and 20 (28%) of 71 patients in the placebo and abiraterone group had grade 3 or worse adverse events, including anaemia (in 15 [21%] of 71 patients vs none of 71), pneumonia (four [6%] vs three [4%]), and myocardial infarction (four [6%] vs none). Serious adverse events were reported by 24 (34%) of 71 patients receiving olaparib and abiraterone (seven of which were related to treatment) and 13 (18%) of 71 patients receiving placebo and abiraterone (one of which was related to treatment). One treatment-related death (pneumonitis) occurred in the olaparib and abiraterone group.

Interpretation Olaparib in combination with abiraterone provided clinical efficacy benefit for patients with metastatic castration-resistant prostate cancer compared with abiraterone alone. More serious adverse events were observed in patients who received olaparib and abiraterone than abiraterone alone. Our data suggest that the combination of olaparib and abiraterone might provide an additional clinical benefit to a broad population of patients with metastatic castration-resistant prostate cancer.

Funding AstraZeneca.

Copyright © 2018 Elsevier Ltd. All rights reserved.

Introduction

Prostate cancer is the fifth leading cause of cancer-associated deaths in men worldwide.¹ The standard of care for metastatic castration-resistant prostate cancer includes taxane chemotherapy (eg, docetaxel or

cabazitaxel), second-generation antihormonal drugs (eg, abiraterone or enzalutamide) that target the androgen-receptor pathway, or radium-223. However, treatment response is often short-lived because patients develop tumour resistance, and thus improved therapeutic

Lancet Oncol 2018; 19: 975–86

Published Online

June 4, 2018

[http://dx.doi.org/10.1016/S1470-2045\(18\)30365-6](http://dx.doi.org/10.1016/S1470-2045(18)30365-6)

See [Comment](#) page 860

The Christie and Salford Royal Hospitals, Manchester, UK (Prof N Clarke ChM); Maria Skłodowska-Curie Memorial Cancer Centre, Warsaw, Poland (P Wiechno MD); Herten Moscow Cancer Research Institute, Moscow, Russia (Prof B Alekseev MD); Catalan Institute of Oncology, Hospital Josep Trueta, Girona, Spain (N Sala MD); Velindre Cancer Centre, Cardiff University, Cardiff, UK (R Jones MRCP); Masaryk Memorial Cancer Institute, Brno, Czech Republic (I Kocak MD); Ospedale Vito Fazzi, Lecce, Italy (V E Chiuri MD); Medical University of Gdańsk, Gdańsk, Poland (Prof J Jassem MD); Centre Léon Bérard, Lyon, France (A Fléchon MD); Sharp HealthCare, San Diego, CA, USA (C Redfern MD); AstraZeneca, Gaithersburg, MD, USA (C Goessl MD, J Burgents PhD); AstraZeneca, Cambridge, UK (R Kozarski PhD, M Learoyd PhD); AstraZeneca, Macclesfield, UK (D Hodgson PhD); and Centre Hospitalier de l'Université de Montréal, Montréal, QC, Canada (Prof F Saad MD FRCS)

Correspondence to: Prof Noel Clarke, Department of Surgery, Christie NHS Foundation Trust, Manchester M20 4BX, UK noel.clarke@christie.nhs.uk

Research in context

Evidence before this study

Olaparib is a poly(ADP-ribose) polymerase (PARP) inhibitor that is approved in ovarian and breast cancer indications. Other PARP inhibitors in clinical development include niraparib, pamiparib, rucaparib, talazoparib, and veliparib. Abiraterone is an androgen synthesis inhibitor, which is the standard of care for metastatic castration-resistant prostate cancer. We searched PubMed and the databases of the American Society of Clinical Oncology and European Society for Medical Oncology for journal publications and meeting abstracts published between Jan 1, 2012, and March 1, 2018, using the search terms "poly(ADP-ribose) polymerase" or "PARP" and "inhibitor" or "inhibition" and "prostate cancer". No language restrictions were applied. In a previous phase 2 study, the efficacy of olaparib monotherapy in patients with advanced prostate cancer was almost exclusively limited to patients with a homologous recombination repair mutation. A phase 2 trial of veliparib in combination with abiraterone in patients with metastatic castration-resistant prostate cancer found no significant efficacy benefit with this combination treatment when compared with abiraterone alone.

Added value of this study

To our knowledge, our data are the first to show a significant improvement in radiographic progression-free survival (rPFS) for men with metastatic castrate-resistant prostate cancer treated with the combination of a PARP inhibitor and androgen synthesis inhibitor compared with an androgen synthesis inhibitor alone. Additionally, and in contrast to the results of previous studies of olaparib monotherapy for this indication, the rPFS benefit was observed for patients with metastatic castration-resistant

prostate cancer given olaparib and abiraterone, regardless of homologous recombination repair mutation status, which has not previously been reported. In our study, more patients treated with olaparib and abiraterone had grade 3 or worse adverse events and serious adverse events than those treated with abiraterone alone; however, the increased duration of exposure in the olaparib and abiraterone group suggests that this tolerability risk might be offset by the observed efficacy benefit. No detriment to health-related quality of life was recorded with the combination compared with abiraterone alone.

Implications of all the available evidence

The significant improvement in rPFS observed in patients with metastatic castration-resistant prostate cancer treated with olaparib plus abiraterone compared with abiraterone alone suggests that these patients, who were not selected on the basis of biomarker criteria, might benefit from the combination treatment irrespective of homologous recombination repair mutation status. This result, which suggests that the treatment could provide clinical benefit for a broader patient population, is consistent with preclinical data that indicate a synergy between olaparib and drugs inhibiting androgen synthesis or function, potentially caused by PARP inhibition of androgen-receptor-dependent transcription or creation of a phenotype similar to that of the *BRCA*ness phenotype that is susceptible to PARP inhibition. Larger studies are needed to confirm the results of this trial; however, our data suggest that the combination of olaparib and abiraterone has the potential to provide additional and practice-changing therapeutic options for men with metastatic castration-resistant prostate cancer.

options are needed for men with metastatic castration-resistant prostate cancer.²

In a preliminary phase 2 study,³ patients with metastatic castration-resistant prostate cancer who were pretreated with chemotherapy (most of whom had also previously received a second-generation antihormonal drug such as abiraterone or enzalutamide) were given the poly(ADP-ribose) polymerase (PARP) inhibitor olaparib. Treatment response was markedly improved in patients with tumours carrying a homologous recombination repair (HRR) mutation (based on a 113-gene panel test) compared with patients with tumours without an HRR mutation.³ These clinical data are supported by preclinical studies investigating the mechanism of action of olaparib, which was found to trap PARP at sites of DNA damage, causing an accumulation of DNA double-strand breaks.⁴ Synthetic lethality is observed when PARP is trapped in HRR-deficient cells, which depend on low-fidelity pathways for repairing DNA double-strand breaks.^{5,6}

Preclinical data^{7,8} have indicated synergy between olaparib and drugs that affect the androgen receptor pathway, regardless of HRR mutation status. Therefore,

we carried out this phase 2 randomised trial to assess the efficacy and tolerability of olaparib in combination with abiraterone compared with placebo plus abiraterone in patients with metastatic castration-resistant prostate cancer, irrespective of their HRR mutation status.

Methods

Study design and participants

We carried out this double-blind, randomised, placebo-controlled, phase 2 trial at 41 urological oncology sites in 11 countries across Europe and North America (appendix p 1). The trial comprised an open-label safety run-in phase, followed by a randomised, double-blind treatment phase.

Eligible male patients were aged 18 years or older with histologically or cytologically proven metastatic castration-resistant prostate cancer. Metastatic castration-resistant prostate cancer was defined as an increasing prostate-specific antigen concentration or other signs of disease progression despite androgen-deprivation therapy and serum testosterone levels at castrate levels (≤ 50 ng/dL), and at least one metastatic lesion on bone scan, CT scan, or MRI. Patients were not required have an HRR

See Online for appendix

mutation. Patients had to be candidates for abiraterone therapy, and had to have an Eastern Cooperative Oncology Group (ECOG) performance status of 0–2 with no deterioration observed in the 2 weeks before the study, and a life expectancy of 12 weeks or longer. For the randomised phase, patients were also required to have received previous treatment with docetaxel for metastatic castration-resistant prostate cancer, but response to this treatment was not necessary. Patients were excluded if they had received more than two previous lines of chemotherapy or had previous exposure to second-generation antihormonal drugs or any previous treatment with olaparib. Patients with abnormal organ, bone marrow, or cardiac function at baseline were excluded on the basis of criteria shown in the appendix (p 3). Patients diagnosed with other malignancies up to 5 years before trial entry, and those with any evidence of severe or uncontrolled systemic diseases, including hypertension or infection, or spinal cord compression or brain metastases (unless asymptomatic and stable) were also excluded. Full inclusion and exclusion criteria are shown in the appendix (pp 69–72). All patients provided written informed consent, and the study protocol was approved by the institutional review board or ethics committee at all participating institutions. The trial was done in accordance with the Declaration of Helsinki, Good Clinical Practice guidelines, and the AstraZeneca policy on bioethics.⁹ No further analyses of the primary outcome measure are planned. The study protocol is available in the appendix (p 12).

Randomisation and masking

Patients were enrolled by the investigators at each individual site and were randomly assigned (1:1) to receive either olaparib plus abiraterone or placebo plus abiraterone using a centralised interactive voice or web response system, without stratification. Investigators contacted the centralised interactive voice response system by telephone or online for allocation of the randomised treatment Kit ID number, which were assigned sequentially to each patient as they became eligible. Patients, those giving the interventions, data collectors, and study personnel were masked to treatment allocation. Individuals involved in data analysis remained masked to treatment allocation until the time of the primary analysis and all investigators and patients remained masked until verification and closure of the study database, with the exception of medical emergencies in which knowledge about treatment group was required for appropriate patient management.

Procedures

We did an open-label run-in phase (appendix pp 2–3) to determine a suitable and safe olaparib dose before the randomised treatment phase. In brief, in this run-in dose-escalation phase, an initial cohort of up to six patients was given an initial dose of olaparib 200 mg

twice daily (plus abiraterone 1000 mg once daily). If this was well tolerated (≤ 1 dose-limiting toxicity in these patients), the olaparib dose was increased to 300 mg twice daily. If this higher dose was also well tolerated (< 4 dose-limiting toxicities in 12 patients), then olaparib 300 mg twice daily was to be used in the next phase of the study. For the randomised treatment phase, patients received oral olaparib tablets 300 mg twice daily with doses taken approximately 12 h apart, plus oral abiraterone once daily in the morning, or matching placebo plus oral abiraterone 1000 mg. Both treatment groups received oral prednisone or prednisolone 5 mg twice daily in addition to abiraterone, with doses taken approximately 12 h apart.¹⁰ Treatment was continued until disease progression or lack of clinical benefit (investigator-assessed), and patients were permitted to discontinue either olaparib or placebo, or abiraterone individually at the discretion of the investigator. Olaparib or placebo dose interruptions of up to 14 days were permitted at the investigator's discretion to manage any toxicities, and were required for grade 3 or 4 treatment-related adverse events (according to the National Cancer Institute Common Terminology Criteria for Adverse Events [NCI-CTCAE] version 4.0). Treatment was reinitiated when adverse events had resolved to grade 1 or less. Patients were considered for dose reduction of olaparib or placebo to 250 mg twice daily and to 200 mg twice daily if toxicity recurred. Interruptions or dose reductions of abiraterone or prednisone or prednisolone were not permitted. Patients could withdraw from the study voluntarily or be withdrawn because of severe protocol non-compliance, and patients who could not be contacted following three or more unsuccessful attempts were considered lost to follow-up.

Soft-tissue CT or MRI and bone scans were done every 12 weeks until week 72, then every 24 weeks thereafter until disease progression, death, or withdrawal of consent. Blood samples were obtained for the assessment of circulating tumour cells at baseline, week 4, week 12, and at discontinuation of study treatment. Blood samples for the assessment of prostate-specific antigen concentrations were taken every 4 weeks until week 12 and every 12 weeks thereafter. Adverse events were monitored throughout the study treatment and 30-day follow-up periods and classified according to NCI-CTCAE version 4.0, and included measurements of clinical chemistry and haematology (at baseline, every 4 weeks until week 52, and every 12 weeks from week 60), vital signs (at baseline, every 4 weeks until week 12, and every 12 weeks thereafter), and recording of adverse events and serious adverse events (at baseline, every 4 weeks until week 24, and every 12 weeks thereafter). Health-related quality of life (HRQOL) was assessed every 4 weeks until week 12, and every 12 weeks thereafter using the Functional Assessment of Cancer Therapy–Prostate (FACT-P) questionnaire. A higher FACT-P score represented better HRQOL (range 0–156).

Plasma (mandatory), whole-blood (germline; optional), and archival tumour tissue (optional) samples were tested for deleterious or suspected deleterious (loss-of-function) mutations in 15 HRR genes according to pre-specified American College of Medical Genetics and Genomics criteria.¹¹ Tumour and whole-blood samples were tested first, and subsequently patients with no data (ie, no material provided or technical test failure) were prioritised for plasma analyses (appendix p 4).

Loss-of-function mutations were assessed in *ATM*, *BARD1*, *BRCA1*, *BRCA2*, *BRIP1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCL*, *PALB2*, *RAD51B*, *RAD51C*, *RAD51D*, and *RAD54L* using the Clinical Laboratory Improvement Amendments (CLIA) FoundationOne tumour assay at Foundation Medicine (Cambridge, MA, USA); in *ATM*, *BARD1*, *BRCA1*, *BRCA2*, *BRIP1*, *CHEK2*, *PALB2*, *RAD51C*, and *RAD51D* using the germline CLIA assay at Color Genomics (Burlingame, CA, USA); and in *ATM*, *BARD1*, *BRCA1*, *BRCA2*, *BRIP1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCL*, *PALB2*, *PPP2R2A*, *RAD51B*, *RAD51C*, *RAD51D*, and *RAD54L* using a research use only plasma assay (AZ100) at the AstraZeneca Genomics Laboratory in Cambridge, UK. The plasma assay (circulating tumour DNA [ctDNA] next-generation sequencing analysis) was done on an Illumina NextSeq 500 system using 2×150 high-throughput runs, at the AstraZeneca Genomics Laboratory. The assay is a 112-gene panel using IDT xGen capture probes that covers the full coding sequence of the HRR genes listed above. The targeted panel and unique molecular indices enrichment method were validated using commercial plasma samples harbouring somatic alterations at known allele frequencies. For the ctDNA analyses, the mean input of circulating free DNA was 29 ng (SD 38), the detection limit was 0·5% allele frequency, and the mean depth after deduplication was 2900×.

Outcomes

The primary endpoint of the randomised phase was investigator-assessed radiographic progression-free survival (rPFS), defined as time from randomisation to radiographic progression (based on Response Evaluation Criteria in Solid Tumors [RECIST] version 1.1 for soft-tissue disease or Prostate Cancer Working Group 2 [PCWG2] criteria for bone disease), or death. Secondary endpoints were safety and tolerability; investigator-assessed time to second progression (PFS2, defined as time from randomisation to the progression event [according to local standard clinical practice, including radiographic, symptomatic or prostate-specific antigen progression] following that used for the primary rPFS analysis, or death); overall survival (defined as time from randomisation to death from any cause); the proportion of patients who achieved a radiographic objective response (based on RECIST version 1.1 or PCWG2 criteria) and the duration of this response, and the proportion of patients who achieved a malignant soft tissue objective response (based on RECIST version 1.1);

time to first subsequent anticancer therapy (TFST) and time to second subsequent anticancer therapy (TSST), which were defined as time from randomisation to the first or second subsequent therapy for prostate cancer following discontinuation of olaparib or placebo, or death; confirmed prostate-specific antigen response ($\geq 50\%$ reduction in prostate-specific antigen concentration from baseline, confirmed at the next assessment ≥ 4 weeks later); change in circulating tumour cell count (from ≥ 5 cells per 7·5 mL at baseline to < 5 cells per 7·5 mL at any assessment post-baseline). A key exploratory endpoint was HRQOL (in which a deterioration event was defined as a decrease of ≥ 6 points from baseline FACT-P score).

Exploratory rPFS subgroup analyses by HRR mutation status were predefined. In these analyses, patients with a qualifying loss-of-function mutation in any sample (tumour, plasma, or whole-blood) were classified as having an HRR mutation; patients for whom all test results were negative were classified as HRR wild-type, as long as their results included a valid tumour test; all other patients were classified as HRR partially characterised, including those whose plasma and whole-blood samples both tested negative for HRR mutations, but for whom no valid tumour test result was available.

Statistical analysis

We calculated that we needed to enrol approximately 140 patients (70 per treatment group) in the randomised phase to give 80% power to detect a statistically significant difference between treatment groups at a one-sided significance level of 10%, assuming a true hazard ratio (HR) of 0·65. A hierarchical multiple-testing strategy was prespecified for the primary analysis of rPFS and the key secondary endpoints PFS2 and overall survival. The final analysis was planned to be done after 100 rPFS events had occurred. If statistical significance was shown for rPFS, PFS2 was then compared between the treatment groups. If the null hypothesis of no difference between treatment groups was rejected for PFS2, overall survival was tested as part of the multiple-testing procedure; however, all planned analyses were done, and p values determined outside of the multiple testing strategy should be considered nominal. Efficacy analyses were done in the intention-to-treat population, which included all randomly assigned patients, and safety and tolerability analyses included all patients who received at least one dose of olaparib or placebo. Objective response and soft-tissue response were assessed in patients with measurable disease at baseline, circulating tumour cell counts were assessed in patients with at least five circulating tumour cells per 7·5 mL at baseline, and prostate-specific antigen response was assessed in patients who had a baseline prostate-specific antigen measurement. For the primary endpoint, patients without disease progression or who had disease progression after two or more missed visits

were censored at their last evaluable tumour assessment. HRs and 95% CIs were calculated using a log-rank test with the Breslow method for ties; an HR of less than 1 favoured olaparib. Time-to-event endpoints were measured from randomisation and medians and accompanying 95% CIs were calculated using the Kaplan-Meier method. All statistical analyses were done using SAS version 9.4. Reported p values are two-sided, and we regarded p values of less than 0.05 as significant.

Predefined exploratory subgroup analyses were done for patients with an HRR mutation, wild-type HRR, and partially characterised HRR status. Prespecified sensitivity analyses of rPFS to assess attrition and evaluation-time bias are described in the appendix (p 3). A logistic regression model including treatment as a factor variable was used to analyse objective response; no other variables were included in this model. There was no data monitoring committee for this study.

This trial is registered with ClinicalTrials.gov, number NCT01972217.

Role of the funding source

The study sponsor, AstraZeneca, was involved in the study design, data collection, data analysis, and data interpretation, and gave approval to submit for publication. Merck & Co, Inc, which is co-developing olaparib, provided input into data interpretation. Both AstraZeneca and Merck & Co, Inc, had a role in the writing of the report through funding of medical writing support. All authors had full access to all the data in the study and were involved in the decision to submit for publication. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

The results of the open-label safety run-in phase are described in the appendix (p 3). In brief, no patients reported dose-limiting toxicities during the dose-escalation, and olaparib 300 mg twice daily was established as the recommended phase 2 dose for the randomised phase. Pharmacokinetic analyses showed no obvious drug-drug interactions between olaparib and abiraterone (appendix pp 10–11).

Between Nov 25, 2014, and July 14, 2015, 171 patients were enrolled in the randomised phase and assessed for eligibility, of whom 142 patients were randomly assigned to receive olaparib plus abiraterone (the olaparib group; 71 patients) or placebo plus abiraterone (the placebo group; 71 patients; figure 1). All patients (71 in the olaparib group and 71 in the placebo group) were included in the efficacy and safety analysis sets. The clinical cutoff date for the final analysis was Sept 22, 2017. At data cutoff, seven (10%) of 71 patients were still receiving olaparib plus abiraterone, and eight (11%) of 71 were still receiving placebo plus abiraterone. 43 (61%) of 71 patients in the olaparib group had died, of which

33 (46%) deaths were due to metastatic castration-resistant prostate cancer, four (6%) were due to adverse events, five (7%) were for reasons unrelated to adverse events or the disease under investigation, and one (1%) death was due to other reasons. 45 (63%) of 71 patients in the placebo group had died, of which 37 (52%) deaths were due to metastatic castration-resistant prostate cancer, one (1%) was due to an adverse event, and seven (10%) deaths were due to reasons unrelated to adverse events or the disease under investigation.

Patient demographics and baseline characteristics are reported in table 1. Patients in the olaparib group had a higher median prostate-specific antigen concentration

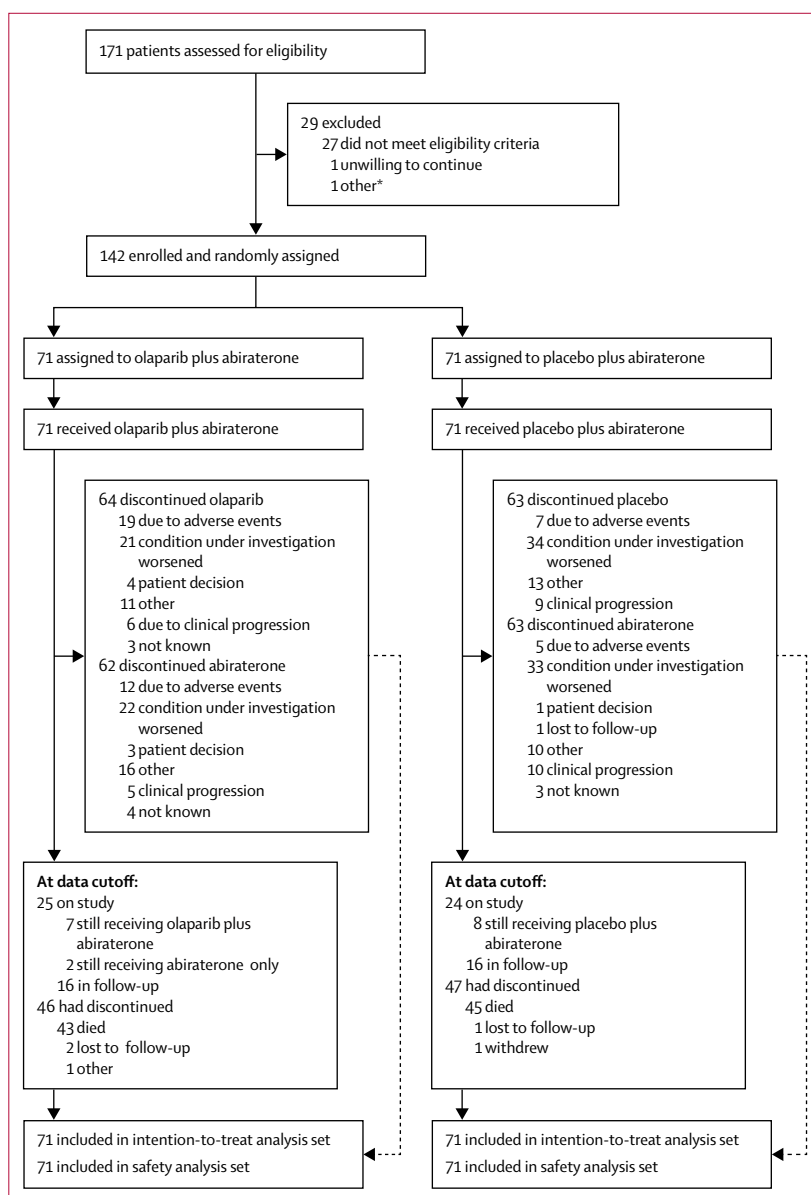


Figure 1: Trial profile

*One patient was incorrectly randomised and did not receive treatment. During data analysis this patient was considered not randomised.

	Olaparib and abiraterone (n=71)	Placebo and abiraterone (n=71)
Age (years)	70 (65–75)	67 (62–74)
Race		
White	67 (94%)	67 (94%)
Black or African-American	1 (1%)	1 (1%)
Asian	1 (1%)	0
Other	2 (3%)	3 (4%)
ECOG performance status		
0	34 (48%)	38 (54%)
1	36 (51%)	30 (42%)
2	1 (1%)	1 (1%)
Unknown	0	2 (3%)
Prostate-specific antigen concentration (µg/L)	86 (23–194)	47 (21–199)
Time from initial diagnosis to first dose (months)	62 (38–93)	48 (32–76)
Extent of disease*		
Bone disease only	33 (46%)	33 (46%)
Soft-tissue disease only	8 (11%)	11 (15%)
Bone and soft-tissue disease	30 (42%)	27 (38%)
Number of bone metastases		
0	5 (7%)	6 (8%)
1	3 (4%)	4 (6%)
2–4	24 (34%)	36 (51%)
5–9	39 (55%)	25 (35%)
HRR mutation status†		
HRR mutation	11 (15%)	10 (14%)
Wild-type HRR	15 (21%)	20 (28%)
HRR partially characterised	45 (63%)	41 (58%)
Previous therapy‡		
Docetaxel	71 (100%)	71 (100%)
Cabazitaxel	10 (14%)	9 (13%)
Abiraterone	0	1 (1%)

Data are n (%) or median (IQR). ECOG=Eastern Cooperative Oncology Group. HRR=homologous recombination repair. *Soft-tissue disease includes visceral organs (liver and lung). †HRR mutation refers to patients with a mutation detected by tumour, germline, or plasma testing; wild-type HRR refers to patients with a valid tumour test result and no HRR mutation detected by any test; HRR partially characterised refers to patients with no HRR mutation detected and no valid tumour test result. ‡One patient in the placebo group had received three previous lines of chemotherapy and one patient had previously received abiraterone; however, these protocol deviations were only reported after randomisation, thus both patients were included in the intention-to-treat analysis set.

Table 1: Patient demographics and baseline characteristics

than did patients in the placebo group. At baseline, 44 (62%) of 71 patients in the olaparib group and 40 (56%) of 71 patients in the placebo group had risk factors for cardiovascular events. 33 (46%) of 71 patients in the olaparib and abiraterone group and 38 (54%) of 71 patients in the placebo and abiraterone group had measurable disease at baseline. Prostate tumour samples were obtained from 68 (48%) of 142 patients (37 in the olaparib group and 31 in the placebo group), 38 (56%) of whom had a valid HRR mutation test result. Tumour, germline,

and plasma test results are shown in the appendix (p 5). 11 (15%) of 71 patients in the olaparib and abiraterone group and ten (14%) of 71 patients in the placebo and abiraterone group had confirmed or suspected deleterious HRR mutations (olaparib and abiraterone group: three patients with *ATM* mutations, two with *BRCA2* mutations, two with *CDK12* mutations, two with *CHEK2* mutations, one with a *BRIP1* mutation, and one with a *CHEK1* mutation; placebo and abiraterone group: four patients with *ATM* mutations, four with *BRCA2* mutations, one with a *CDK12* mutation, and one with a *PALB2* mutation; appendix p 6), and 35 (25%) of 142 patients (15 in the olaparib group and 20 in the placebo group) were classified as HRR wild-type on the basis of their HRR mutation test results, which must have included a negative tumour test result. The remaining 86 (61%) patients (45 [63%] of 71 patients in the olaparib group and 41 [58%] of 71 patients in the placebo group) had partially characterised HRR status, of whom 58 (67%) patients (32 in the olaparib group and 26 in the placebo group) were HRR wild-type by plasma and germline testing (but this could not be verified by tumour testing), 17 (20%) patients (seven in the olaparib group and ten in the placebo group) were HRR wild-type by plasma testing only, five (6%) patients (two in the olaparib group and three in the placebo group) were HRR wild-type by germline testing only, and six (7%) patients (four in the olaparib group and two in the placebo group) had no valid tumour, plasma, or germline test result. Appendix p 5 provides a breakdown of the HRR mutation testing results by the sample type (tumour, germline, or plasma) from which they were obtained. Notably, in the 69 patients with both germline and plasma results, nine patients (13%) were positive for HRR mutations; all were likely to be somatic, and two (22%) of nine were from genes not covered by Color Genomics' germline test (*CDK12* and *CHEK1*). Patients who did not provide a tumour sample or who had no valid tumour test result due to technical failure (excluding the five patients with novel HRR mutations already identified by germline testing) were prioritised for plasma testing.

At the data cutoff for the primary analysis, median follow-up was 15.9 months (IQR 8.1–25.5) in the olaparib and abiraterone group compared with 24.5 months (8.1–27.6) in the placebo and abiraterone group. 46 (65%) of 71 patients in the olaparib group and 54 (76%) of 71 patients in the placebo group had radiographic disease progression events or died. Of the 46 patients in the olaparib group who had radiographic disease progression events or died, 24 (52%) had soft-tissue progression only, eight (17%) had bone progression only, one (2%) had both soft-tissue and bone progression, and 13 (28%) died before progression. Of the 54 patients in the placebo and abiraterone group who had radiographic disease progression events or died, 28 (52%) had soft-tissue progression only, 14 (26%) had bone progression only, two (4%) had soft-tissue and bone

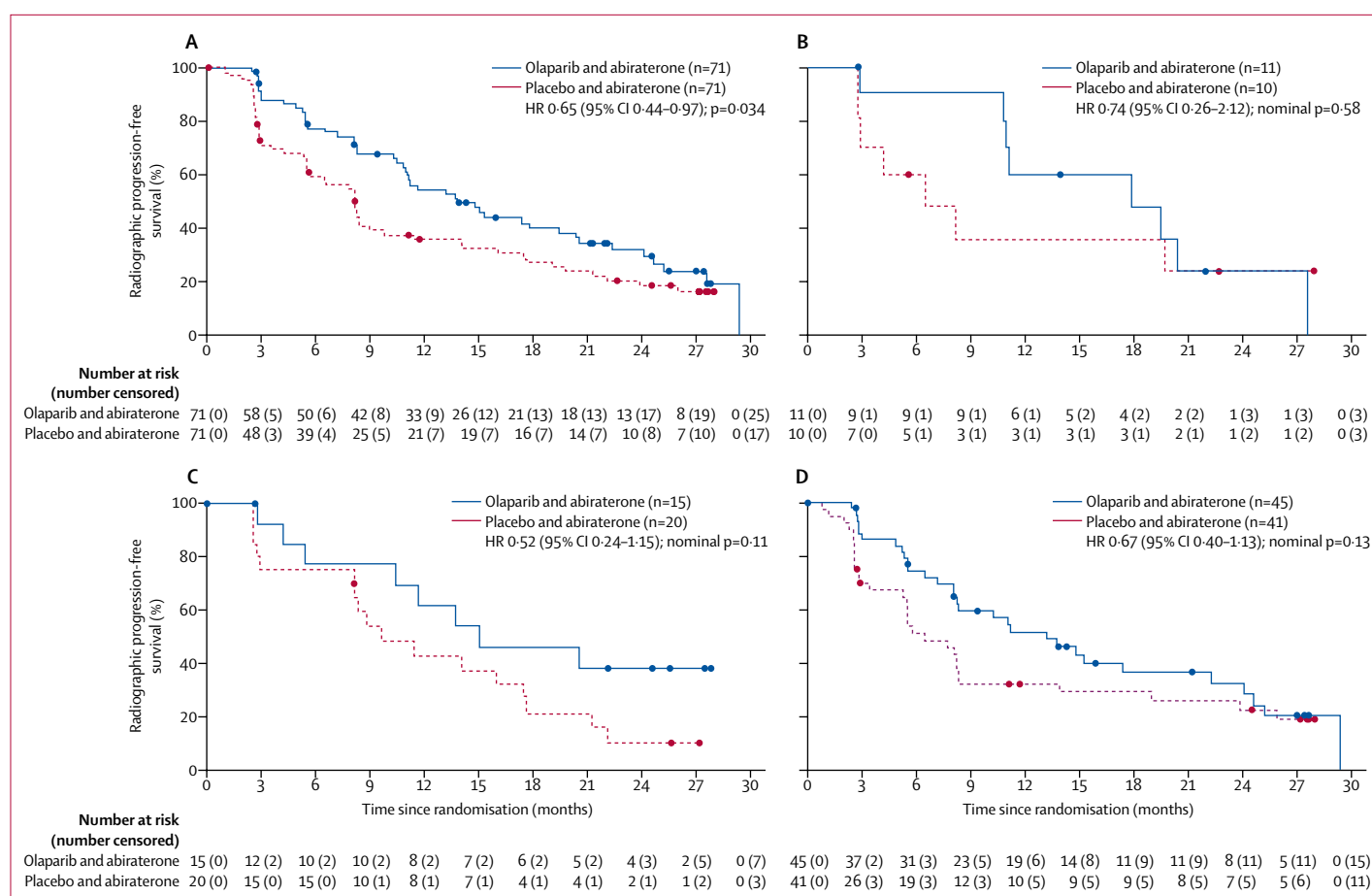


Figure 2: Radiographic progression-free survival in the (A) intention-to-treat population, (B) HRR mutation-positive subgroup, (C) wild-type HRR subgroup, and (D) partially characterised HRR status subgroup

HRR=homologous recombination repair. HR=hazard ratio.

progression, and ten (19%) died before progression. rPFS was significantly longer in the olaparib and abiraterone group than the placebo and abiraterone group (median 13.8 months [95% CI 10.8–20.4] vs 8.2 months [5.5–9.7]; HR 0.65 [95% CI 0.44–0.97], p=0.034; figure 2A). Sensitivity analyses for attrition and evaluation-time bias were consistent with the primary analysis (appendix p 6).

We did prespecified subgroup analyses of rPFS by HRR mutation status. In the 21 patients with HRR mutations, eight (73%) of 11 patients in the olaparib group and seven (70%) of ten patients in the placebo group had radiographic progression or died, and median rPFS was 17.8 months (95% CI 2.9–27.6) in the olaparib group compared with 6.5 months (2.7–not reached) in the placebo group (figure 2B). Of the 35 patients with wild-type HRR, eight (53%) of 15 patients in the olaparib group and 17 (85%) of 20 in the placebo group had radiographic progression or died, with a median rPFS of 15.0 months (95% CI 5.4–not reached) in the olaparib group versus 9.7 months (2.9–17.5) in the placebo group (figure 2C). Of the 86 patients with partially characterised HRR

status, 30 (67%) of 45 patients in the olaparib group and 30 (73%) of 41 in the placebo group had radiographic progression or died, and median rPFS was 13.1 months (95% CI 8.1–22.4) in the olaparib group versus 6.4 months (5.3–8.2) in the placebo group (figure 2D).

By the data cutoff, 37 (52%) of 71 patients in the olaparib group and 45 (63%) of 71 patients in the placebo group had a second progression event or died. Median PFS2 was 23.3 months (95% CI 17.4–not reached) in the olaparib group compared with 18.5 months (16.1–23.8) in the placebo group (figure 3A). At the data cutoff, 43 (61%) of 71 patients in the olaparib group and 45 (63%) of 71 in the placebo group had died. Median overall survival was 22.7 months (95% CI 17.4–29.4) in the olaparib and abiraterone group compared with 20.9 months (17.6–26.3) in the placebo and abiraterone group (figure 3B).

In the patients with measurable disease at baseline (n=33 in the olaparib group; n=38 in the placebo group), no significant difference was identified between the treatment groups in the proportion of patients who achieved an overall objective response (nine [27%] of

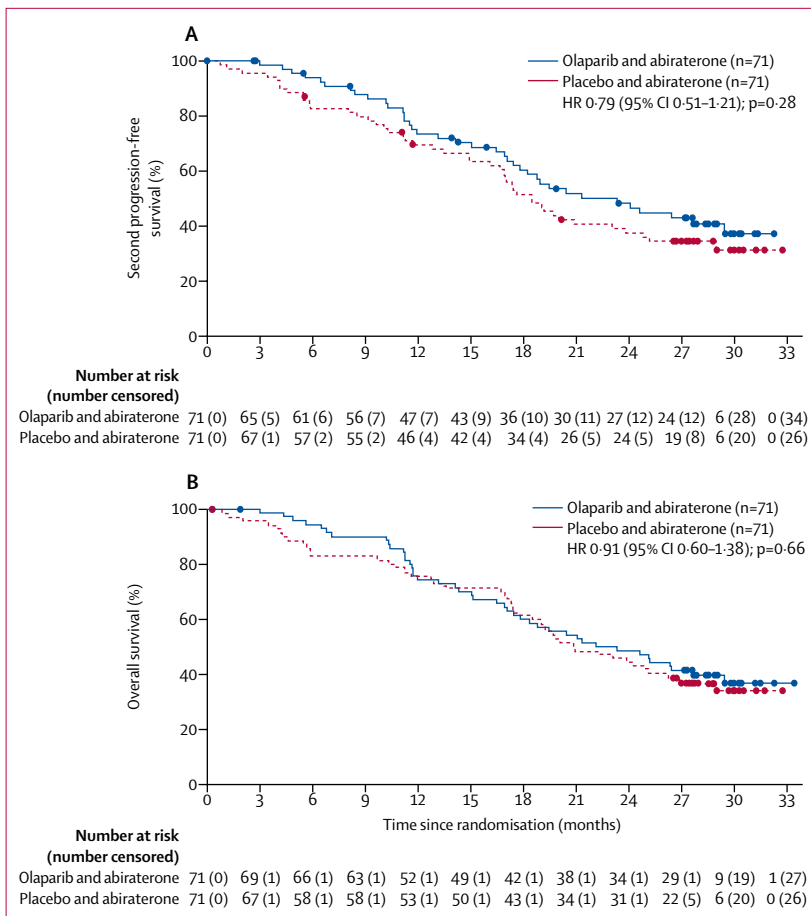


Figure 3: Kaplan-Meier estimates of (A) time to second progression or death and (B) overall survival

33 patients in the olaparib and abiraterone group vs 12 [32%] of 38 patients in the placebo and abiraterone group; odds ratio 0.81, 95% CI 0.28–2.26, nominal $p=0.62$). No patients had a complete response; best response was stable disease in 16 (48%) of 33 patients in the olaparib group and eight (21%) of 38 patients in the placebo group, and progressive disease in seven (21%) patients in the olaparib group and 18 (47%) patients in the placebo group; one patient in the olaparib group was not evaluable because of incomplete post-baseline assessments. Median duration of response was 17.8 months (IQR 8.3–non-calculable) in the olaparib and abiraterone group and 12.1 months (6.6–non-calculable) in the placebo and abiraterone group. Of the patients with a circulating tumour cell count of 5 or more per 7.5 mL at baseline (30 [42%] of 71 patients in the olaparib group and 28 [39%] of 71 patients in the placebo group), 15 (50%) of 30 patients in the olaparib group and 13 (46%) of 28 patients in the placebo group had a decrease in cell count (to <5 cells per 7.5 mL). 34 (48%) of 71 patients in the olaparib and abiraterone group compared with 30 (42%) of 71 patients in the placebo and abiraterone group had a confirmed prostate-specific antigen response.

Soft-tissue responses, and TFST and TSST analyses are reported in the appendix (pp 6–7).

Median treatment duration was 309 days (IQR 145–457) for olaparib compared with 253 days (113–421) for placebo. Median treatment duration with abiraterone was 338 days (IQR 169–588) in the olaparib group versus 253 days (130–429) in the placebo group. During the study, fewer patients in the olaparib and abiraterone group (20 [28%] of 71) than the placebo and abiraterone group (29 [41%] of 71) had anticancer therapy after progression (full details of post-progression therapies are on appendix p 6).

In the olaparib and abiraterone group, 66 (93%) of 71 patients experienced an adverse event, compared with 57 (80%) of 71 patients in the placebo and abiraterone group. Most adverse events in both treatment groups were grade 1 or 2 (table 2). The most common grade 1–2 adverse events were nausea (26 [37%] patients in the olaparib group vs 13 [18%] patients in the placebo group, constipation (18 [25%] vs eight [11%]), and back pain (17 [24%] vs 13 [18%]). More patients had grade 3 or worse adverse events in the olaparib and abiraterone group than in the placebo and abiraterone group (38 [54%] of 71 vs 20 [28%] of 71). Grade 3 or worse adverse events that were reported more frequently in the olaparib group than in the placebo group included anaemia (in 15 [21%] of 71 patients vs none of 71), pneumonia (four [6%] vs three [4%]), and myocardial infarction (four [6%] vs none; table 2).

Serious adverse events were reported by 24 (34%) of 71 patients in the olaparib and abiraterone group and 13 (18%) of 71 patients in the placebo and abiraterone group. Seven (10%) of 71 patients in the olaparib group had serious adverse events that were related to study treatment (anaemia [n=3], febrile neutropenia [n=1], pneumonitis [n=1], vomiting [n=1], general deterioration in physical health [n=1]) compared with one (1%) of 71 patients in the placebo group (gastroenteritis [n=1]). Seven (10%) of 71 patients in the olaparib and abiraterone group had serious cardiovascular events (myocardial infarction [n=4], fatal cardiac failure [n=1], chronic cardiac failure [n=1], fatal ischaemic stroke [n=1]) compared with one patient in the placebo group (thrombotic stroke [n=1]). In the olaparib and abiraterone group, times to onset of myocardial infarction were 5, 9, 24, and 29 months; time to onset of fatal cardiac failure, chronic cardiac failure, and fatal ischaemic stroke were 3, 4, and 18 months, respectively. In the placebo and abiraterone group time to onset of thrombotic stroke was 17 months.

Pneumonitis was reported by two (3%) of 71 patients in the olaparib group and interstitial lung disease was reported in one (1%) of 71 patients in the placebo group. Deaths due to adverse events occurred in four (6%) of 71 patients in the olaparib group (pneumonitis [n=1], ischaemic stroke [n=1], cardiac failure [n=1], mediastinitis [n=1]) and in one (1%) of 71 patients in the placebo group (chronic pyelonephritis [n=1]). Of the four patients who

	Olaparib and abiraterone (n=71)				Placebo and abiraterone (n=71)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
All	28 (39%)	29 (41%)	5 (7%)	4 (6%)	37 (52%)	19 (27%)	0	1 (1%)
Nausea	26 (37%)	1 (1%)	0	0	13 (18%)	2 (3%)	0	0
Constipation	18 (25%)	0	0	0	8 (11%)	0	0	0
Back pain	17 (24%)	1 (1%)	0	0	13 (18%)	1 (1%)	0	0
Fatigue	14 (20%)	1 (1%)	0	0	7 (10%)	2 (3%)	0	0
Asthenia	13 (18%)	3 (4%)	0	0	10 (14%)	0	0	0
Vomiting	13 (18%)	2 (3%)	0	0	8 (11%)	1 (1%)	0	0
Peripheral oedema	13 (18%)	0	0	0	8 (11%)	0	0	0
Decreased appetite	12 (17%)	0	0	0	4 (6%)	1 (1%)	0	0
Diarrhoea	11 (15%)	0	0	0	7 (10%)	1 (1%)	0	0
Dyspnoea	10 (14%)	0	0	0	4 (6%)	1 (1%)	0	0
Pyrexia	10 (14%)	0	0	0	1 (1%)	0	0	0
Cough	9 (13%)	2 (3%)	0	0	2 (3%)	0	0	0
Bone pain	9 (13%)	1 (1%)	0	0	7 (10%)	1 (1%)	0	0
Urinary tract infection	8 (11%)	1 (1%)	0	0	1 (1%)	2 (3%)	0	0
Arthralgia	8 (11%)	0	0	0	3 (4%)	1 (1%)	0	0
Viral upper respiratory tract infection	8 (11%)	0	0	0	3 (4%)	0	0	0
Abdominal pain	8 (11%)	0	0	0	1 (1%)	0	0	0
Anaemia	7 (10%)	14 (20%)	1 (1%)	0	1 (1%)	0	0	0
Neutropenia	7 (10%)	1 (1%)	0	0	0	0	0	0
Hypokalaemia	4 (6%)	2 (3%)	0	0	4 (6%)	0	0	0
Pneumonia	2 (3%)	2 (3%)	2 (3%)	0	0	3 (4%)	0	0
Musculoskeletal chest pain	1 (1%)	0	0	0	3 (4%)	2 (3%)	0	0
Myocardial infarction	0	4 (6%)	0	0	0	0	0	0

Data are n (%). The table shows grade 1–2 adverse events that occurred in 10% or more patients in either group and grade 3–5 events that occurred in 2% or more patients in either group. The full table of adverse events including all grade 3–5 events is in the appendix (pp 8–9).

Table 2: Adverse events

died in the olaparib group, only pneumonitis was considered treatment-related by the investigator; the single death in the placebo group was not considered to be treatment-related.

More patients in the olaparib and abiraterone group than in the placebo and abiraterone group had dose interruptions (24 [34%] of 71 vs nine [13%] of 71) and dose reductions (13 [18%] of 71 vs none of 71) due to adverse events (appendix p 7). 21 (30%) of 71 patients in the olaparib group and seven (10%) of 71 patients in the placebo group discontinued study treatment due to adverse events. The most common adverse events that led to treatment discontinuation in the olaparib and abiraterone group were anaemia (n=4), nausea (n=3), muscular weakness (n=2), and myocardial infarction (n=2); in the placebo and abiraterone group the most common adverse event that led to treatment discontinuation was vomiting (n=3).

In our prespecified exploratory analysis of HRQOL, 60 (85%) of 71 patients in the olaparib and abiraterone group had deterioration in HRQOL (a decrease in FACT-P score of ≥ 6 points from baseline) compared with 57 (80%) of 71 patients in the placebo and abiraterone group; median time to deterioration was 5.7 months (95% CI 2.8–11.2) versus 6.0 months (1.9–11.2),

respectively (HR 0.97, 95% CI 0.68–1.40, nominal $p=0.89$).

Discussion

The results of this randomised phase 2 study show that the addition of olaparib to abiraterone therapy results in a significant rPFS benefit for patients with metastatic castration-resistant prostate cancer, with a significantly longer rPFS in the olaparib and abiraterone group than in the placebo and abiraterone group. Although patients given olaparib had more adverse events than did those who received placebo, median treatment duration was longer for patients treated with olaparib, and combined therapy did not lead to a decline in HRQOL. To our knowledge, this is the first study of a PARP inhibitor to show a clinical benefit when combined with abiraterone for patients with metastatic castration-resistant prostate cancer who have previously received docetaxel.

Notably, the median rPFS in the placebo and abiraterone group (8.2 months) was longer than the median rPFS of 5.6 months reported in the active arm of a phase 3 trial, COU-AA-301,² which assessed abiraterone compared with placebo in a similar patient population. One possible explanation for this difference in median rPFS in the abiraterone group between these two trials is

the difference in tumour load; patients in the previously published study had a higher median prostate-specific antigen concentration at baseline than did the patients in our study. Another potential reason for the difference observed is the improvement in treatment methods and increased access to second-line and third-line interventions for the patients in our study.

Our primary endpoint was predicated on preclinical data showing synergy between olaparib and drugs that affect the androgen receptor pathway, regardless of HRR mutation status; therefore, HRR mutation status was not used as a stratification factor at randomisation.^{7,8} This study was not powered for subgroup analyses, and HRR mutation status was not known for all patients. However, with rPFS results indicative of similar benefit with the combination of olaparib and abiraterone across different HRR subgroups, our data suggest that the drug combination might have resulted in rPFS benefit for patients regardless of HRR mutation status. 61% of patients were classified as having partially characterised HRR mutation status according to our prespecified criteria, but 67% of these patients tested negative for HRR mutations by both plasma and germline testing, although these results were not confirmed by tumour testing. Moreover, because plasma testing was prioritised for patients who did not provide a tumour sample or who had no valid tumour test result, concordance comparisons between germline, plasma, and tumour testing for HRR mutations are intrinsically biased; however, we plan to address this in a future publication.

Previous studies^{3,12,13} have shown that up to 30% of patients with metastatic castration-resistant prostate cancer have HRR mutations. In our trial, the prevalence of detected HRR mutation was approximately 15% (appendix pp 4–5). Within the cohort tested at Color Genomics, 93 of 102 patients also had a tumour or plasma result, and seven of the 16 mutations detected were germline, consistent with published data.^{3,12,13} All samples in our study were from primary prostate tumours; however, a matched study of primary and metastatic tumour samples from patients with metastatic castration-resistant prostate cancer found only a small number of additional HRR mutations in metastatic samples,¹⁴ suggesting the type of sample used in our study is unlikely to have had much of an effect on the prevalence of detected HRR mutations. Additionally, *BRCA2* mutations have been associated with worse prognosis in metastatic castration-resistant prostate cancer,¹⁵ which might explain their increased prevalence in patients with metastatic castration-resistant prostate cancer.^{15,16} One limitation of our study was that several patients with a partially characterised HRR status might have had an HRR mutation. According to previous studies that have indicated that 30% of patients with metastatic castration-resistant prostate cancer have HRR mutations, it is possible that up to 13 patients with a partially characterised HRR status actually had an HRR

mutation, which would increase the total number of patients with a HRR mutation but no valid tumour test to 31 patients. However, the actual number is likely to be much lower, in view of the fact that 63 of 86 patients with a partially characterised HRR status did not have a germline HRR mutation according to Color Genomics testing, and 75 of 86 patients did not have a detectable mutation in their plasma (which would be expected to detect all germline variants and point mutations in patients with tumours that had shed sufficient ctDNA). Thus, we would expect the number of patients in the partially characterised subgroup with an HRR mutation to be less than 13 and this group is unlikely to be driving the treatment effect observed in the subgroup of patients with partially characterised HRR status or in the study as a whole. Abiraterone (irrespective of PARP inhibitor use) might be more effective in patients with an HRR mutation than in those without; however, in our study the number of patients with a known HRR mutation was similar between the treatment groups, thus any effect of HRR mutation status on abiraterone efficacy is also likely to be well balanced.

In the TOPARP-A study,³ the efficacy of olaparib monotherapy in patients with metastatic castration-resistant prostate cancer was almost completely limited to patients with HRR mutations, whereas our data suggest that synergy between olaparib and abiraterone might result in clinical benefit for patients regardless of HRR mutation status. Preclinical studies suggest a dual mode of synergy. First, PARP is involved in androgen receptor-dependent transcription and PARP inhibition impairs this process.⁸ Second, the androgen receptor regulates transcription of DNA repair genes; androgen depletion impairs HRR, which might produce a so-called *BRCAness* phenotype that is susceptible to PARP inhibition.^{17–19} More studies are required to elucidate the mechanism further. However, a broad patient population might derive benefit from combined treatment with PARP inhibitors and drugs that induce a *BRCAness* phenotype. In a phase 2 study¹² with a similar sample size (n=148) that investigated a different PARP inhibitor—veliparib—in combination with abiraterone compared with abiraterone alone, no significant efficacy benefit was observed for patients with metastatic castration-resistant prostate cancer, including those with DNA damage repair defects. However, these results are consistent with preclinical data²⁰ that show veliparib to have weaker PARP trapping activity than olaparib.

Although no statistically significant difference in PFS2 or overall survival was found between the treatment groups, our study was not powered for these analyses. The overall survival analysis might have also been confounded by the higher proportion of patients in the placebo and abiraterone group than the olaparib and abiraterone group who received anticancer therapies after progression, and some differences between the treatment groups in baseline prognostic factors such as age, ECOG

performance status, and prostate-specific antigen concentration. However, because this was a relatively small phase 2 trial, no stratification factors were applied during randomisation. Although no significant difference in PFS2 was found between the treatment groups, the magnitude and direction of the PFS2 data were consistent with the primary rPFS analysis, and the difference in median overall survival between treatment groups (1·8 months) is similar to that reported for available second-line therapies.²¹ However, the number of patients in these analyses is too small to draw conclusions and further studies are needed to determine whether or not the observed rPFS benefit translates into improved overall survival. Overall response, confirmed prostate-specific antigen response, and circulating tumour cell conversion rates were similar in both treatment groups, which might be a result of the potency of abiraterone and its cytostatic mode of action.^{2,22} The observed difference in rPFS between treatment groups is therefore likely to be due to an increase in the proportion of patients with stable disease and an increased duration of response in the olaparib and abiraterone group, rather than the result of an increased number of responders. This notion is consistent with the proposed mechanism of action for the combination treatment, in which a *BRCAness* phenotype is only established in patients for whom abiraterone has efficacy.^{7,18}

The most frequent adverse events of any grade in the olaparib group were nausea and anaemia. More patients in the olaparib and abiraterone group had grade 3 or worse adverse events, serious adverse events, and other clinically significant adverse events, including cardiovascular events, than patients in the placebo group. No difference in risk factors for cardiovascular events were identified between the treatment groups at baseline. Additionally, more patients in the olaparib and abiraterone group than the placebo and abiraterone group had a dose modification or discontinued treatment because of an adverse event. In the olaparib and abiraterone group, patients were permitted to discontinue one or both treatments, resulting in a longer median duration of treatment for abiraterone than olaparib in this group. However, the longer duration of both treatments in the olaparib and abiraterone treatment group than in the placebo group suggests any risk of decreased tolerability might be offset by the additional efficacy benefit and the increase in rPFS with the combination was observed despite more treatment discontinuations due to adverse events. No differences in HRQOL were identified between the two groups, which might be explained by the additional disease burden experienced by patients in the placebo and abiraterone group and the natural history of the disease at this late stage.

Our study was limited by the number of patients enrolled, and larger trials are needed to confirm and extend our observations. It should also be noted that patients with metastatic castration-resistant prostate

cancer who have received docetaxel, but no second-generation antihormonal drugs, might be increasingly rare, making recruitment and appropriate statistical power in this setting a challenge and increasing the importance of assessing this combination in other metastatic castration-resistant prostate cancer settings. Assessment of the effect of previous docetaxel treatment on the efficacy of olaparib and abiraterone will be an important area for future research and, since previous studies have shown significant increases in overall survival for patients naive to hormonal therapy or chemotherapy treated with abiraterone, testing the combination of a PARP inhibitor and abiraterone in pre-chemotherapy settings might be of particular interest.^{23,24} An ongoing phase 2 trial (NCT03012321) in the pre-chemotherapy metastatic castration-resistant prostate cancer setting is investigating olaparib versus abiraterone versus olaparib combined with abiraterone in patients with DNA repair defects.²⁵ Olaparib is also currently being studied as monotherapy in a phase 3 trial (NCT02987543) of patients with metastatic castration-resistant prostate cancer with an HRR mutation who have previously received treatment with an antihormonal drug.²⁶ Our results are encouraging for the future development of drug combinations of PARP inhibitors and antihormonal agents, and open up the possibility of larger trials in patients with metastatic castration-resistant prostate cancer, irrespective of HRR mutation status.

In conclusion, our data provide evidence of clinical benefit for men with metastatic castration-resistant prostate cancer given olaparib in combination with abiraterone compared with abiraterone alone, and indicate that this combination could potentially benefit patients, regardless of HRR mutation status.

Contributors

NC was responsible for study design and writing of the manuscript. NC, PW, BA, NS, RJ, IK, VEC, JJ, AF, CR, and FS were responsible for patient accrual, trial conduct, and obtaining the data. CG, JB, RK, DH, and ML analysed the data. All authors interpreted the data and reviewed the draft and final versions of the manuscript.

Declaration of interests

BA reports grants, personal fees, and travel expenses from AstraZeneca, Janssen, Pfizer, Merck & Co, Roche, and Sanofi. NS reports personal fees from Astellas Pharma, Bristol-Myers Squibb, and Janssen. VEC reports personal fees and travel expenses from Bristol-Myers Squibb, Janssen, and Pfizer. JJ reports personal fees from AstraZeneca. AF has received honoraria and travel expenses from Astellas Pharma, AstraZeneca, Bristol-Myers Squibb, Janssen, Pfizer, and Sanofi. CG, JB, DH, and ML are employees of AstraZeneca and own stock. RK was employed by AstraZeneca during manuscript development. FS reports grants and personal fees from Astellas Pharma, AstraZeneca, and Janssen during the conduct of the study; and grants and personal fees from Sanofi and Bayer outside the submitted work. All other authors declare no competing interests.

Acknowledgments

We thank all the patients who participated in the study and their families, and our co-investigators. This study was sponsored by AstraZeneca and coordinated by IQVIA. Medical writing support was provided by Rachel Patel and Elin Pyke (Mudskipper Business Ltd, Bollington, UK) and was funded by AstraZeneca and Merck & Co.

References

- 1 Ferlay J, Soerjomataram I, Ervik M et al. GLOBOCAN 2012 v1.1, cancer incidence and mortality worldwide: IARC CancerBase No. 11. Lyon, France: International Agency for Research on Cancer, 2014.
- 2 de Bono JS, Logothetis CJ, Molina A, et al. Abiraterone and increased survival in metastatic prostate cancer. *N Engl J Med* 2011; **364**: 1995–2005.
- 3 Mateo J, Carreira S, Sandhu S, et al. DNA-repair defects and olaparib in metastatic prostate cancer. *N Engl J Med* 2015; **373**: 1697–708.
- 4 Murai J, Huang SY, Das BB, et al. Trapping of PARP1 and PARP2 by clinical PARP inhibitors. *Cancer Res* 2012; **72**: 5588–99.
- 5 Farmer H, McCabe N, Lord CJ, et al. Targeting the DNA repair defect in BRCA mutant cells as a therapeutic strategy. *Nature* 2005; **434**: 917–21.
- 6 Bryant HE, Schultz N, Thomas HD, et al. Specific killing of BRCA2-deficient tumors with inhibition of poly(ADP-ribose) polymerase. *Nature* 2005; **434**: 913–17.
- 7 Asim M, Tarish F, Zecchini HI, et al. Synthetic lethality between androgen receptor signalling and the PARP pathway in prostate cancer. *Nat Commun* 2017; **8**: 374.
- 8 Schiewer MJ, Goodwin JF, Han S, et al. Dual roles of PARP-1 promote cancer growth and progression. *Cancer Discov* 2012; **2**: 1134–49.
- 9 AstraZeneca. Global policy. 2016. https://www.astrazeneca.com/content/dam/az/PDF/Bioethics_policy.pdf (accessed July 8, 2016).
- 10 Janssen. ZYTIGA (abiraterone acetate) prescribing information. 2017. <http://www.janssenlabels.com/package-insert/product-monograph/prescribing-information/ZYTIGA-pi.pdf> (accessed May 3, 2018).
- 11 Richards S, Aziz N, Bale S, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genet Med* 2015; **17**: 405–24.
- 12 Hussain M, Daignault-Newton S, Twardowski PW, et al. Targeting androgen receptor and DNA repair in metastatic castration-resistant prostate cancer: results from NCI 9012. *J Clin Oncol* 2018; **36**: 991–99.
- 13 Robinson D, Van Allen EM, Wu YM, et al. Integrative clinical genomics of advanced prostate cancer. *Cell* 2015; **161**: 1215–28.
- 14 Mateo J, Carreira S, Seed G et al. Targeted sequencing for molecular stratification of matched primary tumor samples and metastatic biopsies in castration-resistant prostate cancer. 2016. https://www.pcf.org/wp-content/uploads/2016/10/MATEO_JOAQUIN_19388090_ABSTRACT.pdf (accessed May 2, 2018; abstr).
- 15 Castro E, Goh C, Olmos D, et al. Germline BRCA mutations are associated with higher risk of nodal involvement, distant metastasis, and poor survival outcomes in prostate cancer. *J Clin Oncol* 2013; **31**: 1748–57.
- 16 Castro E, Goh C, Leongamornlert D, et al. Effect of BRCA mutations on metastatic relapse and cause-specific survival after radical treatment for localised prostate cancer. *Eur Urol* 2014; **68**: 186–93.
- 17 Bartek J, Mistrik M, Bartkova J. Androgen receptor signaling fuels DNA repair and radioresistance in prostate cancer. *Cancer Discov* 2013; **3**: 1222–24.
- 18 Polkinghorne WR, Parker JS, Lee MX, et al. Androgen receptor signaling regulates DNA repair in prostate cancers. *Cancer Discov* 2013; **3**: 1245–53.
- 19 Li L, Karanika S, Yang G, et al. Androgen receptor inhibitor-induced “BRCAness” and PARP inhibition are synthetically lethal for castration-resistant prostate cancer. *Sci Signal* 2017; **10**: pii: eaam7479.
- 20 Pommier Y, O'Connor MJ, de Bono J. Laying a trap to kill cancer cells: PARP inhibitors and their mechanisms of action. *Sci Transl Med* 2016; **8**: 362ps17.
- 21 Bahl A, Masson S, Birtle A, Chowdhury S, de Bono J. Second-line treatment options in metastatic castration-resistant prostate cancer: a comparison of key trials with recently approved agents. *Cancer Treat Rev* 2014; **40**: 170–77.
- 22 Attard G, Reid AH, A'Hern R, et al. Selective inhibition of CYP17 with abiraterone acetate is highly active in the treatment of castration-resistant prostate cancer. *J Clin Oncol* 2009; **27**: 3742–48.
- 23 Fizazi K, Tran N, Fein L, et al. Abiraterone plus prednisone in metastatic, castration-sensitive prostate cancer. *N Engl J Med* 2017; **377**: 352–60.
- 24 James ND, de Bono JS, Spears MR, et al. Abiraterone for prostate cancer not previously treated with hormone therapy. *N Engl J Med* 2017; **377**: 338–51.
- 25 Reichert Z, Carneiro BA, Daignault-Newton S, et al. A randomized phase II trial of abiraterone, olaparib or abiraterone + olaparib in patients with metastatic castration-resistant prostate cancer with DNA repair defects. *Proc Am Soc Clin Oncol* 2017; **35**: TPS5086.
- 26 de Bono JS, Hussain M, Thiery-Vuillemin A, et al. PROfound: a randomized phase III trial evaluating olaparib in patients with metastatic castration-resistant prostate cancer and a deleterious homologous recombination DNA repair aberration. *Proc Am Soc Clin Oncol* 2017; **35**: TPS5091.